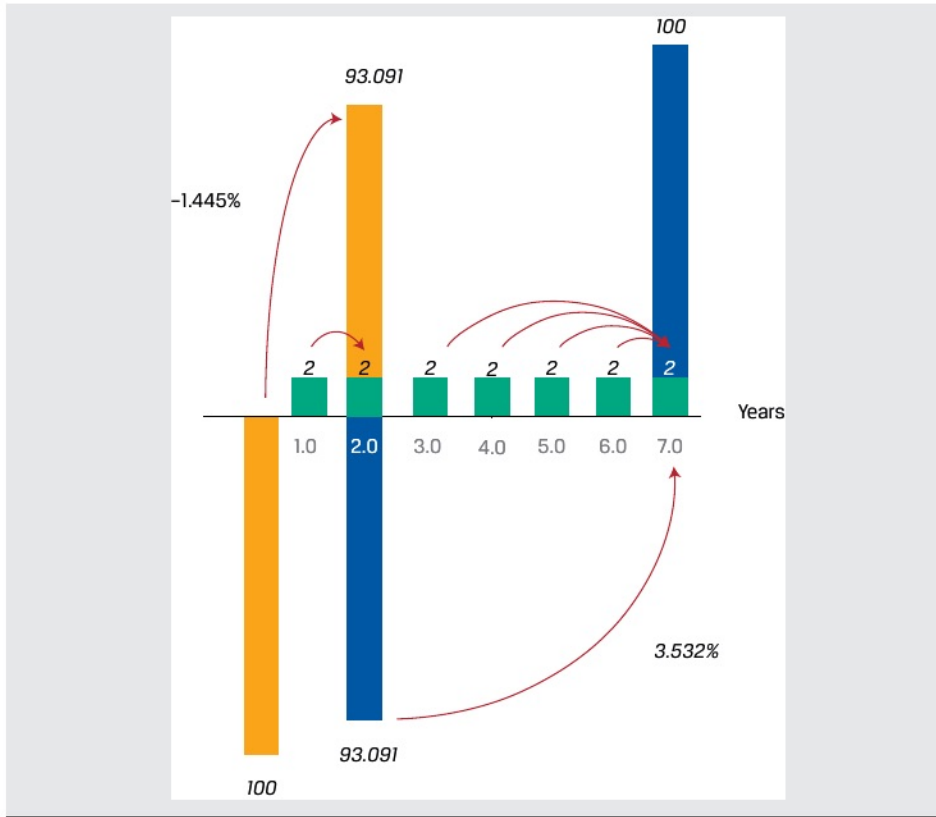




Equity Instruments, Implied Return, and Implied Growth

As noted in the discussion of calculating the present value of an equity investment, the price of a share of stock reflects not only the required return but also the growth of cash flows. If we begin with an assumption of constant growth of dividends from Equation 14, we can rearrange the formula as follows:

$$r - g = \frac{D_t(1 + g)}{PV_t} = \frac{D_{t+1}}{PV_t} \quad (20)$$

The left-hand side of Equation 20 simply reflects the difference between the required return and the constant growth rate, and the right-hand side is the dividend yield of the stock based on expected dividends over the next period. Thus, the implied return on a stock given its expected dividend yield and growth is given by Equation 21, as follows:

$$r = \frac{D_t(1 + g)}{PV_t} + g = \frac{D_{t+1}}{PV_t} + g \quad (21)$$

Simply put, if we assume that a stock's dividend grows at a constant rate in perpetuity, the stock's implied return is equal to its expected dividend yield plus the constant growth rate.

Alternatively, we may be interested in solving for a stock's implied growth rate, and this relation is given by Equation 22:

$$g = \frac{r * PV_t - D_t}{PV_t + D_t} = r - \frac{D_{t+1}}{PV_t} \quad (22)$$

If a stock's next expected dividend is known, then we can calculate the implied growth by deducting its expected dividend yield from its required return.